

Course Name: Exercise 1

Power method for finding dominant eigenvector of a matrix

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0.1 Extracting the eigenvalue of a known eigenvector

Let $\vec{v} \in \mathbb{R}^n$ be an eigenvector of the matrix $\mathbf{A}$ with the corresponding eigenvalue λ . Then by normalizing $\vec{v}$ and multiplying it with $\mathbf{A}$ from both the right and the left, we can extract λ :

$$\begin{aligned}\hat{\mathbf{v}}^T \mathbf{A} \hat{\mathbf{v}} &= \hat{\mathbf{v}}^T \lambda \hat{\mathbf{v}} \\ &= \lambda \hat{\mathbf{v}}^T \hat{\mathbf{v}} \\ &= \lambda \|\hat{\mathbf{v}}\|^2 \\ &= \lambda \cdot 1 \\ &= \lambda.\end{aligned}\tag{1}$$

0.2 The power method

The *dominant eigenvalue* λ_{dom} of a matrix $\mathbf{A}$ of dimensions $n \times n$ is the eigenvalue with largest (absolute) value:

$$\lambda_{\text{dom}} = \max |\lambda_i|,\tag{2}$$

where $\{\lambda_i\}$ are the eigenvalues of $\mathbf{A}$.

One can find the dominant eigenvector of $\mathbf{A}$ by using the *power method*:

1. let $\vec{v}_0 \in \mathbb{R}^n$ be a **non-zero** vector with random components.
2. Set $\lambda_0 = 1$ (note: the subscript denotes an iteration step, not the order of the eigenvalue).
3. Apply the $\mathbf{A}$ to $\vec{v}_0$: $\vec{v}_1 = \mathbf{A}\vec{v}_0$.
4. Extract λ_1 by multiplying $\mathbf{A}$ from the left by $\vec{v}_1^T$ and from the right by $\vec{v}_1$ (??).